Problem 27.3

A perpetuity pays 1,000 at the beginning of the first year. Each subsequent payment is increased by inflation. If inflation is assumed to be 4% per year, calculate the present value of this perpetuity using an annual effective interest rate of 4%.

Solution

Index the payments by $n = 0, 1, 2, \ldots$. The payment at time n is $1{,}000(1.04)^n$, while the annual effective interest rate is also 4%. Thus, the present value is

$$\begin{aligned} PV &= \sum_{n=0}^{\infty} \frac{1{,}000(1.04)^n}{(1.04)^n} \\ &= \sum_{n=0}^{\infty} 1{,}000. \end{aligned}$$

Each payment has a present value of 1,000, so the sum diverges and there is no finite present value. Equivalently, the growing-perpetuity formula

$$PV = \frac{1{,}000}{1 - \frac{1+g}{1+i}}$$

requires $i > g$. Here $i = g = 0.04$, making the denominator zero.

Therefore, the perpetuity has no finite present value:

$$\boxed{\text{The present value diverges.}}$$